EC4308 Project Guide

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Aims and scope.

The main goal of this project is to get a feel of what it means to apply the techniques learned in this course to prediction problems in economics and finance. Some may view it as an opportunity to work out methodology and data for your upcoming honors thesis. Due to time constraints, it is not expected that students produce enough output to constitute a standalone research paper. Instead, please be **moderate and realistic** in outlining the scope of your project. A compact but well-thought-out research statement would be preferable. For example, it may be too cumbersome to satisfactorily extend existing academic machine learning research or use/construct previously unavailable data – these are best left for your future research such as a thesis. However, it might be a good idea to play with a particular part of a research paper or methodology that you are interested in and try a few simple tweaks – e.g., try several variations of a method to see relative performance, or a combination of several methods, or reapply the analysis in an existing study to another sufficiently rich and freely available dataset. Again, the main goal is to obtain thorough hands-on experience applying ML methods learned in class and those similar to them rather than pushing the boundaries of research. Therefore, **students are expected to spend the most effort to provide as deep as possible an analysis using tools mentioned in class in a more "standard" setting**, rather than spend a lot of time on learning extra-curricular methods and constructing/cleaning data while providing a relatively shallow forecasting exercise.

Therefore, the formulation of a project topic should go along the following lines:

1) Determine what your primary interest is – is it micro, macro or financial prediction problem? What variable(s) do you want to forecast? – different approaches may be appropriate for different variables depending on their type (continuous vs. categorical), cross-sectional vs. time series data etc.
2) Examine available data for the variable you are going to forecast, and any other variables you may use as predictors. Given your interest and data available, think about what would be interesting (and sensible) to do with it using the methods we learned and methods you may have learned outside of class. E.g., a good starting point is checking

whether some previous finding holds for a different setting (different country, in a different time period etc.). If there is no suitable data available – you may want to change the research question in a way that it becomes feasible. Remember – for us here **data is everything**. Do not try to squeeze out something from a small dataset just because it is from a country/market of some special interest to you – go for richer, perhaps more standard datasets that either accompany papers or are available from reputable sources (more on this below). Remember, you would need a suitably large training sample for estimating your model, a validation sample in some cases, and then a sizeable testing sample for out-of-sample evaluation. At the same time, ML does not equal Big Data, so you don't need to obsess about finding huge datasets with thousands of observations and predictors. Remember, through simple feature engineering you can blow up your number of potential predictors quite considerably if necessary.

3) Given the course's focus on R, you shall use it to complete your projects. Some authors provide replication code online (including R packages!) for implementing advanced procedures – keep that in mind, it may well let you delve into more interesting methods. **To keep your research reproducible and peer-reviewable, do keep track of and document appropriately all the operations you do in your code, particularly data processing!** Don't forget to set the seed(s) whenever random draws are involved!
4) One way to split up work among group members would be to first collectively work out a methodology, e.g. write a workhorse R script that can be used to obtain results from every method considered. Then, split up the actual estimation, formatting, and interpretation of results among group members. Finally, compile the project together and iron out the kinks – make sure everyone agrees on what is being said in the interpretation so that there are no inconsistencies in the report.

Finding data online.

These days students are extremely lucky since there are tons of economic and financial data available online. You may not always find what you want, but you will always find something to work with. Below I list some common types of places to look for data, which work for economic data in general.

1) Government agencies (e.g., Statistics Offices) and Central Banks' websites. Government agencies and central banks typically provide a lot of data on their respective country. You can expect to find data on GDP components, population, interest rates, government bond prices etc. It may not be always obvious that the data is there sometimes, but pay attention and you'll be able to dig down to it.
2) Specialized databases (e.g., Datastream, Bloomberg, Wharton Research Databases etc.). These are usually restricted in access to particular departments/staff even within the university subscribed to them, but if you can get access, you can get relevant data.
3) **Data from journal articles**. Many journals require the authors to submit original code and data for the article to be published. The data is then available for the readers to download. This comes in handy when replicating empirical findings or using same data for other purposes. Some relevant journals for us are Econometrica, Journal of Applied Econometrics (every paper in JAE is required to disclose data and code, unless protected by an NDA), Journal of Business and Economic Statistics etc. If the article you find interesting was recently published in a decent journal, chances are data is available in the supplementary material (unless data is proprietary) or on the authors' personal webpage(s).
4) **Economists' personal webpages**. Many professors working in applied economic research in general and on ML application in particular, post datasets on their websites. Don't forget to check the authors' webpages of the papers you find interesting (at least in terms of data).
5) Special interest websites like Yahoo! Finance and Bloomberg can provide useful data as well.
6) It is also useful to visit textbook websites. Many (like, e.g., our favorite Stock and Watson) post datasets online.
7) Don't forget about R itself! Some of its packages include datafiles that serve as examples, e.g., the ISLR and MASS packages we used many times in class. Some of those could contain interesting and relevant data.
8) Finally, Google. If it is there and available, it can be found, so never give up without a thorough search!
9) I also have some (including high-frequency) financial data that may be useful – ask me if you are struggling to find frequent data on some financial variables of interest for your project (such as stocks, crypto, futures or foreign exchange).

I compile a list of useful links (that's in no way exhaustive) that should get you started:

Data search engines and collections of links:

https://data.gov.sg/ - a recently launched Singapore government data portal.

https://www.kaggle.com/datasets - datasets used in forecasting/data mining competitions. Many are garbage, but some interesting data can be found nonetheless.

Journal of Applied Econometrics Data Archive (data and sometimes code for all the articles published there):
https://journaldata.zbw.eu/journals/jae

FRED economic database (many series on US and other countries):
https://research.stlouisfed.org/fred2/

FRED-MD and FRED-QD databases – the Big Data of macro. The website has a helpful working paper explaining the data and its properties. We will work with this dataset in class when we talk about time series
https://research.stlouisfed.org/econ/mccracken/fred-databases

UK and Canada versions of monthly macro databases:
https://www.stevanovic.uqam.ca/DS_UKMD.html
https://www.stevanovic.uqam.ca/DS_LCMD.html

Links to some journal article data archives:
http://www.aeaweb.org/RFE/showCat.php?cat_id=9

Researching articles

While for this project you are not required to do an extensive literature review or for that matter use methods that are necessarily outside the course material, it is sometimes helpful to search around to see what people have been doing in the area you are interested in, and also to potentially identify the article you want to borrow data from. We are particularly lucky as NUS boasts subscriptions to almost every possible economic database and journal that can be remotely useful. The main tools you could use would be ScienceDirect and Scopus (all available on the library website under databases). You should probably start with Scopus, as it is a very flexible abstract search tool with very wide coverage that can point you to the articles that are relevant to your search. You can then locate and download the article. I provide some sample articles on Canvas.

Producing a report

Although the scope of this project is limited, I would ask you nevertheless to follow the guidelines for writing a short research article. That means your report should have a particular structure and not be contaminated by academically inappropriate language. View this as an opportunity to practice writing research reports. The structure of the report is usually as follows.

First, you have the introduction, which should motivate what you are doing and explain why it is interesting to the reader. Also, you should mention a couple of key papers in the area in relation to the topic if appropriate (if you were writing a thesis – here would come your literature review) and how what you are doing is expanding on existing knowledge (even a tiny contribution is perfectly acceptable, there are almost no low hanging fruit left in economic research!). At the end, give a brief summary of your results and explain how the rest of the report is organized. Remember – the two parts of the report that are usually read by people are the introduction and conclusion. Based on those, a person should be able to understand broadly what you are doing and whether it is interesting for him/her. Nobody wants to read what's in-between without good reason, so having a good well thought-out introduction is paramount, or else your report

will end up in the bin even if it does not deserve to (I am generally referring to journal referees or job superiors here). This is why you write the introduction section of your report LAST! After you know everything that you've done and the results are there – only then can you start writing an engaging introduction. One of my PhD classmates spent 2 months on his introduction to maximize the impact of an already complete excellent paper!

After the introduction you would want to add a section on methodology where you describe what methods are used in your study and why. Be brief but very clear about what and how you do so that your peers would be able to understand and replicate what you did after reading your report. This could also be a good place to detail any R packages that you are using especially if we haven't seen them in class.

Afterwards, it is customary to discuss the data. Be very clear about the source, any transformations/cleaning you applied to the data, and what is the sample period/size, including how you split into training/validation/test setts. It may be useful to plot the data and examine some simple descriptive statistics here just to give a taste of what the data is like and to potentially spot outliers/things that don't make sense. For time series data, it would help spot trends, seasonality, obvious breaks etc.

Finally, you may start on the main body of your work – reporting and discussing the results of your analysis. Try to connect your analysis to your research question and to general economic knowledge that you have acquired over the years. If there are things that seem out of place – focus attention on it and try to think why it happens. In short, try to make your discussion a bit deeper than "we can see that the line in Figure 1 is upward sloping". Label your figures and tables. Add legends and write notes if necessary so that the reader can understand what's going on in the figure/table without reading the text preceding it (it is customary to explain the variables, abbreviations, cases considered etc. in the note to the figure/table).

Finally, write some concluding remarks. Summarize what you have done and whether the results were as expected. Highlight the interesting points you encountered along the way. Remember – a negative result is also a valuable result that tells us that something doesn't work under certain circumstances, so don't despair if you find that the

performance of whatever method you chose was not as expected. Finally, comment on the limitations of your study (there always are) and possible future research in the area – it is an important indicator of how well you understand what you can and cannot do with the tools and data you have.

Project deliverables

To keep things under control, I would require each group to **email me a 1-page project proposal by the end of Week 7 (Sunday, 23.59pm), but feel free to submit as early as you want**. By that time, we would have hopefully covered a lot of the core material in the class necessary to start on the project. The proposal should contain: a brief outline of the idea, as well as data and methods you plan to use. This is to make sure everyone has a feasible target and is not doing something that's too difficult, too easy, plain impossible or nonsensical.

The project report, not to exceed 15 pages in 11pt font with 1.5 line spacing, will be **due in Week 13 (provisionally Monday).** Note that 15 pages is NOT a target and it's perfectly fine to produce, e.g., a 12-page report – quality rather than quantity is what's important. If you feel like you are breaching the limit (e.g., you have done so much good stuff that it physically doesn't fit without cropping important results), you may inquire with me for a possible exception. The soft copy of the report is to be submitted together with all the data and code to ensure replicability of results (you will put data and code into one zip-file submit on Canvas). The code should be as clear and as detailed as possible (e.g., if you manipulate data inside R, all the variable transformations etc. should be documented in your script), and should run given the raw data file provided.

We shall then listen to group presentations during the Week 13 class. Groups will prepare a 10-minute presentation explaining the goal of their project and what they managed to learn. **All group members should participate in the presentation** (e.g., you can split it up into general intro, data and methods, results, conclusions and further issues).

Project analysis components

To make sure your report is comprehensive, it may be useful to follow the checklist below.

- Discuss **briefly** the approaches to prediction that you are taking. Does the data seem to possess the properties that the methods would work well with? How would you select between competing methods? If there are important/sensible benchmark forecasts to beat in the context of your research question? – be sure to use them and compare the results with your more sophisticated ones. Do not waste space explaining methods we learnt in class in detail, however, do elaborate on ideas of methods outside class material if you are using those.

- Perform the training/test dataset split at the start of your project. Do not use the test data in model estimation/selection in any way (resist the urge to "fish" for a model based on test sample performance). Pretend you do not know these data at all until you are done selecting your models/methods. Invoke them only when you are evaluating the forecast. This makes the exercise more honest. It's perfectly ok if things do not work well – that's life. Correct execution is more important than the result here. Also note that the need to reserve the data for out-of-sample forecast evaluation places additional demands on the total sample size you need to have – hence, avoid small datasets as the data would be cut even shorter by reserving the out-of-sample period!

- Construct additional features from your data, if appropriate, and briefly explain the rationale.

- Train your model(s) on the training set. Explain the training process and any notable points that arise.

- If you are working with time-series data, you should consider an empirically relevant set of forecast horizons, not just 1-step. In many situations, 1-step forecast is not so interesting, but h-step is the main focus. E.g., for monthly macro data, one could examine 1-step, 3-step, 6-step, 12-step forecasts; for daily financial data, one could look at 1-day, 5-day (weekly), 20 or 22-day (monthly) horizons.

- Perform out-of-sample evaluation. Report the usual statistics. Is another loss function (not MSE) more appropriate in your case? (e.g., direction-of-change) – perform evaluation using that loss function as well and discuss your findings.

- If you are considering several candidate methods, you should also examine the performance of the ensemble method combining the chosen models.

- Perform tests for equal predictive ability to see whether differences in forecasting performance are actually significant.

- Provide a holistic discussion of your results in the context of the methodology, existing literature (did other studies tend to find similar results?) as well as your prior expectations (did the most complex methods provide best performance? Were some of the initial hunches you had correct?).

Potential topic-starters

Here I collect a few topics and some papers to help you jump-start project ideas. Some of these also have data available so you do not have to worry about cleaning etc. if you choose to use the datasets. Also, you may try to search for these papers in Scopus, then click on citations to see which papers cited them – this way you can find what developments/improvements may have been implemented.

1) **Macroeconomic forecasting using many predictors.** Prediction in economics typically concerns macroeconomic time series. Here, a large set of predictors at monthly frequency can be considered from the FRED-MD, UK-MD or Canada-MD databases to forecast. You can choose a variable that interests you, e.g., inflation or GDP growth (proxied by industrial activity/production). You can compare how ML methods perform vs. an established benchmark (e.g., random walk for inflation, AR(p) model for GDP growth etc.). There is a nice paper on inflation forecasting by Medeiros et al. (2019) using the FRED-MD data and providing R codes, which we will partially

replicate in one of the lectures. Other macro variables could be forecasted using a similar setup. Coulombe et al. (2021) is an example of a similar study for UK, which also provides ideas on feature engineering. Coulombe et al. (2022) provides some general discussion on ML methods in macro forecasting. UK and Canada data available here: https://www.stevanovic.uqam.ca/index.html
Difficulty: easy-to-medium.

2) **Predictive regressions in finance**: The literature on this is understandably large. A good starting point is a paper by Goyal and Welch (2008) who scrutinize many traditional methods. Their data (and updated version too) is available at Prof. Goyal's webpage: https://sites.google.com/view/agoyal145 Also check Ivo Welch's page: http://www.ivo-welch.info/professional/goyal-welch/ Goyal et al. (2024) provide a more recent follow-up and dataset. Gu et al. (2020) is an influential study looking at relative performance of machine learning methods in explaining equity premia. Neely et al. (2014) can be consulted on getting ideas for extra predictors based on technical analysis. Another issue here to consider is what should be the criterion for ranking the forecasts, as minimizing mean squared error is not necessarily the goal of financial forecasters. Difficulty: easy-to-advanced.

3) **Categorical prediction** (recessions, bull/bear markets etc.). It is interesting to see whether ML methods can beat a simple probit/logit modelling augmented with dynamic features. The influential paper is Kauppi and Saikonnen (2008). Other papers in the zip file show more applications, which can be improved upon with ML methods.
Difficulty: easy-to-challenging.

This is, of course, neither an exhaustive list of topics/papers, nor a list of things that you must absolutely consider. I just tried to provide a few starting points for your search that are established studies and for which data is relatively freely available. You are, of course, free to formulate your own topic. Do not worry too much about implementing the most advanced techniques. The main idea is to utilize **the full potential** of the **techniques you learn in this class** in analyzing a dataset **comprehensively** and seeing how your work stacks up against a

well-established benchmark. Projects using fancy methods but lacking in depth of analysis (e.g., only 1-step forecasts considered in time series, no tests of predictive ability etc.) will not earn good grades. **Of course, you are welcome to discuss your project idea with me and clarify any guidelines and details at any point!**

AI Use Policy

This is a brief project-specific clarification of acceptable, unacceptable and transparent use of generative AI tools (e.g., ChatGPT, Claude, Gemini etc.) in your group project. You should also consult the NUS AI guidelines at:
https://libguides.nus.edu.sg/new2nus/ai_guidelines_infographics

1. Acceptable Uses

- Light editing or proofreading of writing (grammar, spelling, clarity). I recommend to not rely too much on AI in this respect and keep your own voice and style.

- Brainstorming / minor assistance (e.g., phrasing suggestions, code syntax checks).

2. Unacceptable Uses

- Core project content: do not use AI to generate motivation, interpret results, or write discussion. These are pertaining to the key learning goals in this course.

- Bulk text/code generation: do not ask AI to produce very large parts of your report or entire scripts.

- Reference management: do not rely on AI to generate bibliographic references or citations.

Example of an acceptable prompt: *Why does my code *inserted* fail with the error message 'object not found'?*

Comment – here you let GenAI help you identify the problem, and provide some suggestions how to fix the syntax. Since these are often due to typos or

Example of an unacceptable prompt: *Let Y be the data matrix with the first column being the target variable and the rest predictors. Write an R function that performs rolling window 1-step-ahead forecasting of the target variable using Random Forest.*

Comment – here you sort of "outsource your brain" to the GenAI and potentially compromise the learning goals of the class. By letting GenAI complete a rather large section of code by itself, you miss the opportunity to understand how the forecasting logic works by foregoing coding the structure of the operation by yourself.

3. Referencing AI Use

Transparency: acknowledge all AI use, keep chats organized (e.g., within a single project), provide either links to chats or exported transcripts documenting usage.

4. Doubts and Clarifications

Whenever you are unsure whether your intended use of AI fits within these guidelines, consult me before proceeding. There is no penalty for acceptable, declared use. I reserve the right to ask for elaboration on code or writing that appear to have been AI-generated without proper referencing.